

Exit Ticket Questions for Week 3

1. Why can't a rational expression $X(z)$ by itself uniquely determine $x[n]$?
2. Why does a zero on the unit circle create a notch filter?

Good answers:

1. Because different ROCs correspond to different convergent power-series expansions and therefore different time-domain supports.

Look for left/right-sided power series expansions

2. Because at the notch frequency ω_0 the point $e^{j\omega_0}$ coincides with the zero (z_0), making a numerator distance zero and therefore $|H(e^{j\omega_0})| = 0$.

Look for something that shows $|e^{j\omega} - z_0| = 0$ as $\omega \rightarrow \omega_0$

A good exit ticket answer should be short, correct, mathematically sound, to the point and demonstrate a well-reasoned understanding of the subject that the question is asking about. Exit ticket answers will be graded on a scale of “low” (1 point), “medium” (2 points), or “high” (3 points).

Use LaTeX style for math expressions, e.g., $\delta[n]$ for a discrete-time impulse, or use the math equations in the rich content editor in Canvas.